



الرسم بالحاسب

9:00-11:00

26/1/2023

ا.د/ خالد فتحي

Faculty of Computers & Information, Assiut University

Course code: (Add the course code)

3rd Level

Final Exam

Duration: 2 hours

Total Marks: 50

Reviewer: (Add the name)

This exam for the following program(s) : (Add the program's name(s))

Hi, Mahmoud. When you submit this form, the owner will see your name and email address.

* Required

1

* الإسم الرباعي (بالعربي فقط)

Enter your answer

2

* رقم الجلوس

Enter your answer

3

* المستوي

- ☐ الاول
- ☐ الثاني
- ☐ الثالث
- ☐ الرابع 2013
- ☐ الرابع 2014
- ☐ الرابع 2015
- ☐ الرابع 2016
- ☐ الرابع 2017
- ☐ الرابع 2018
- ☐ الرابع 2019

4

* البرنامج

- ☐ بايو
- ☐ هندسة
- ☐ ذكاء صناعي

5

* رقم المعمل

Select your answer



6

* رقم الكمبيوتر

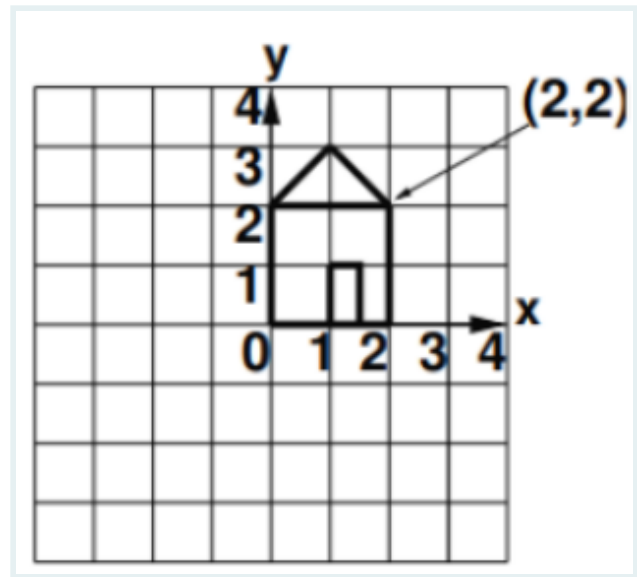
Enter your answer

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

Enter your answer

8



Assume that drawHouse function draws the following house

```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0); glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glScalef(-1.0, -1.0,1.0);
glRotate(90, 0, 0, 1);
glRotate(90, 0, 0, 1);
drawHouse();
glPopMatrix();
```

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:

(1 Point)

- ☐ (0,-2)
- ☐ (2,-2)
- ☐ (-2,2)
- ☐ (-2,-2)

☐ (2,2)

9

The Phong reflectance model does not consider the geometry of the object being shaded

(1 Point)

☐ True

X

☒ False

10

Does OpenGL use column-major order or row-major order when representing a transformation matrix?

(1 Point)

☐ row-major

☒ column-major

11

OpenGL maintains a set of error flags. Each flag represents a different type of error. Whenever one of these errors occurs, the corresponding flag is set. To see whether any of these flags is set, you should call glGetError

(1 Point)

☒ True

☐ False

12

The process of removal of hidden surfaces is termed as

(1 Point)

- ☐ clipping
- ☐ copying
- ☐ culling
- ☐ shorting

13

glShaderSource() hands shader source code to the shader object so that it can keep a copy of it

(1 Point)

- ☒ True
- ☐ False

14

What does glMaterialf(...) do?

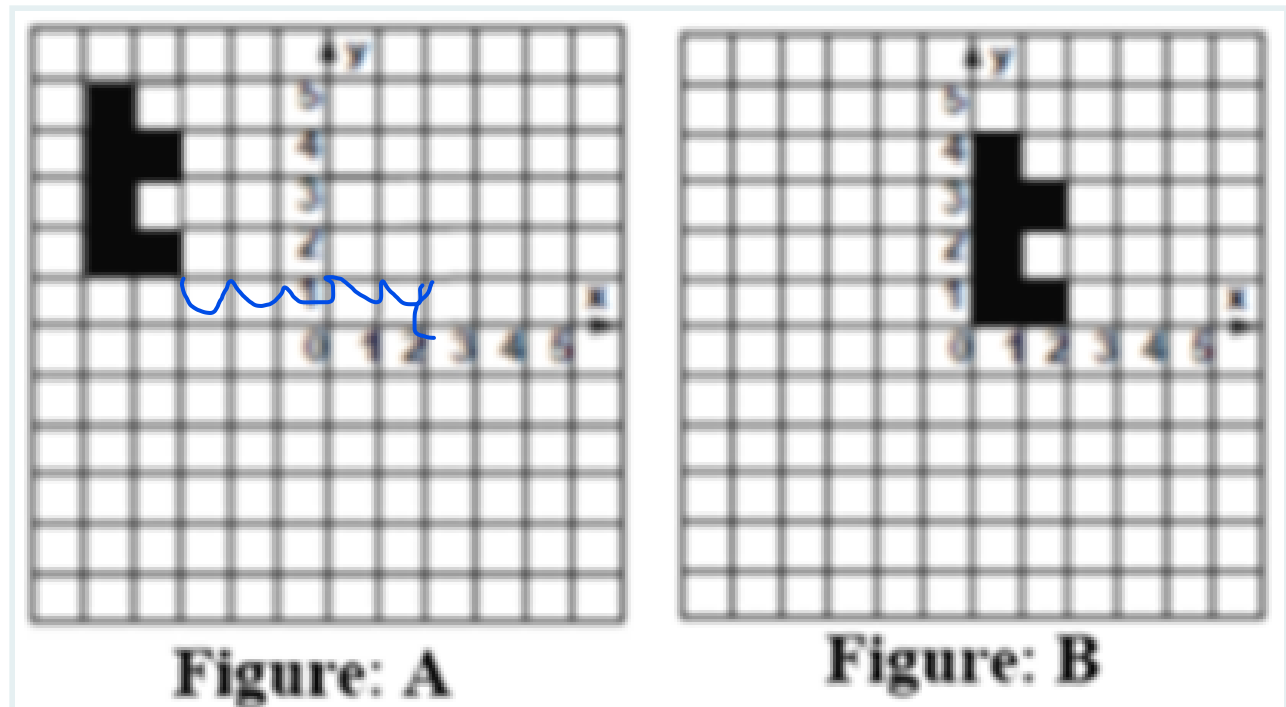
(1 Point)

- ☐ Turns material properties on
- ☐ Switches to material mode to add visual effects
- ☐ Gets the current material properties of a drawn object
- ☒ Defines the material properties of a drawn object for visual effects

15

The OpenGL command(s) that makes the transformation of Figure A to Figure B:

(1 Point)



- ☐ `glTranslatef(5.0, -1.0, 0.0);`
- ☐ `glTranslatef(-5.0, 1.0, 0.0);`
- ☐ `glRotatef(90.0, 0.0, 0.0, 1.0); glTranslatef(-1.0, 1.0, 0.0);`
- ☐ `glTranslatef(-1.0, 1.0, 0.0); glRotatef(90.0, 0.0, 0.0, 1.0);`

16

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glScalef(-2.0, 2.0, 1.0);
```

```
glRotatef(90.0, 0, 0, 1.0);
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 2 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☐ C

☒ D

17

OpenGL can draw convex polygons.

(1 Point)

☒ True

☐ False

18

The transformation matrix that corresponds to the following OpenGL code segment is:

```
glRotatef(90.0, 1.0, 0.0, 0.0);
glTranslatef(-2.0, 1.0, -3.0);
glScalef(1.0, 2.0, 2.0);
(1 Point)
```

A)	B)	C)	D)
$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 0 & 2 & 3 \\ 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 0 & -2 & 3 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 0 & -2 & 1 \\ 0 & 2 & 0 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & -2 \\ 0 & 0 & 2 & 3 \\ 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☒ B

☐ C

☐ D

19

If the GL_SPOT_CUTOFF is 180, spotlight effect equals

(1 Point)

☐ 0

☒ 1

☐ 2

☐ 3

20

For a rotation matrix R and a translation matrix T , $RT = TR$, even though matrix multiplication is not generally commutative

(1 Point)

☐ True

☒ False

21

In OpenGL, the argument of _____ determines which object is to draw.

(1 Point)

☐ glEnd()

☒ glBegin()

☐ glStart()

☐ None of these

22

How do you activate a shader program in OpenGL?

(1 Point)

- ☐ glUseProgram(ProgramID);
- ☒ glUseProgram(ProgramID);
- ☐ glProgram(ProgramID);
- ☐ glShader(ProgramID);
- ☐ glUseShader(ProgramID);

23

$$\begin{bmatrix} 0 & -1 & 0 & 4 \\ 1 & 0 & 0 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The sequence of OpenGL transformations that would produce the following transformation matrix is:

(1 Point)

- ☐ glRotatef(90, 0, 0, 1); glTranslatef(4,-3,2);
- ☒ glTranslatef(4,-3,2); glRotatef(90, 0, 0, 1);
- ☐ glRotatef(90, 0, 0, 1); glScalef(-1.0, 1.0,1.0); glTranslatef(4,-3,2);
- ☐ glTranslatef(4,-3,2); glScalef(-1.0, 1.0,1.0); glRotatef(90, 0, 0, 1);

24



Suppose that you have 10 triangles that can be arranged into a triangle strip. How many vertices would you have to specify to OpenGL if you use the fact that they can be arranged into a triangle strip?

(1 Point)

☐ 9

☐ 10

☐ 11

☒ 12

☐ 13

$$\begin{aligned} V &= n + 2 \\ &= 10 + 2 = 12 \end{aligned}$$

25

Generally, what primitive polygon is used for creating a mesh to represent a complex object?

(1 Point)

☒ Triangle

☐ Cube

☐ Sphere

☐ Circle

26

Which of the following equation is used in 2D translation to move a point(x,y) to the new point (x',y')?

(1 Point)

- ☐ $x' = x + ty$ and $y' = y + tx$
- ☐ $x' = x * tx$ and $y' = y * ty$
- ☒ $x' = x + tx$ and $y' = y + ty$
- ☐ $x' = x + tx$ and $y' = y - ty$

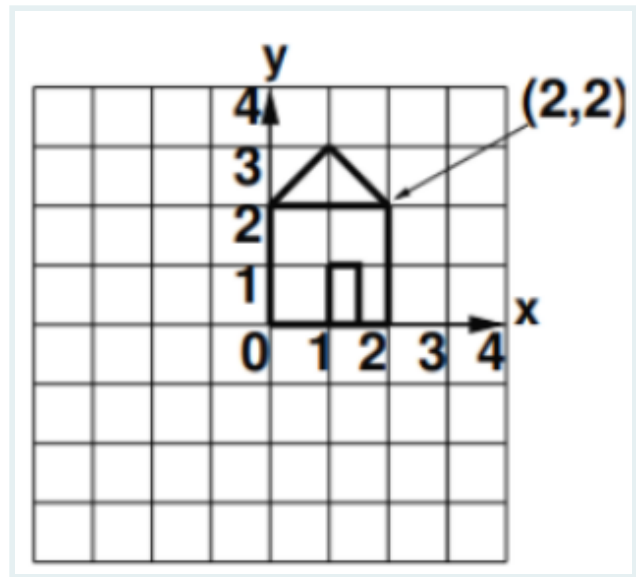
27

The RGB of A cyan color is

(1 Point)

- ☐ (1.0,1.0,0.0)
- ☐ (1.0,0.0,1.0)
- ☒ (0.0,1.0,1.0)
- ☐ (1.0,1.0,1.0)

28



Assume that drawHouse function draws the following house

```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0); glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glRotate(-90, 0, 0, 1);
drawHouse();
glPopMatrix();
```

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:

(1 Point)

☐ (0,-2)

☒ (2,-2)

☐ (-2,2)

☐ (-2,-2)

☐ (2,2)



29

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(90.0, 0, 0, 1.0);
glRotatef(90.0, 0, 0, 1.0);
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☐ C

☒ D

30

The fragment shader determines the position of the vertices

(1 Point)

☐ True

☒ False

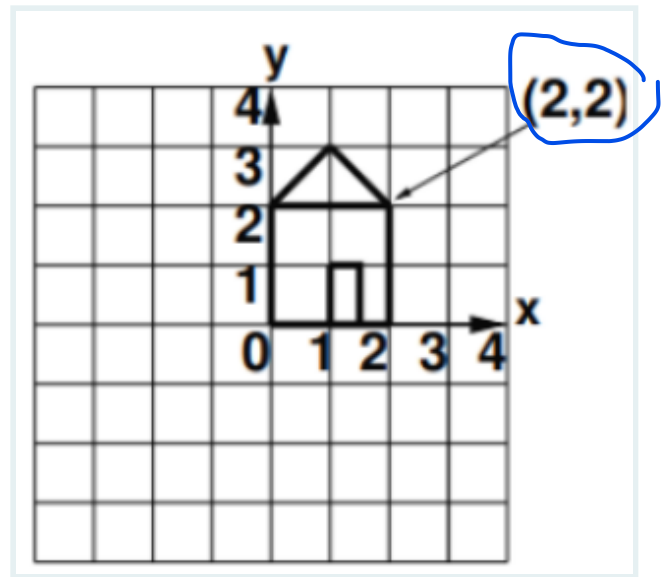
31

What are two types of shaders?

(1 Point)

- ☐ Vertex and polygon.
- ☐ Fragment and polygon.
- ☒ Vertex and fragment.

32



```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
```

```
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0); glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glPushMatrix();
```

```
glScalef(-1.0, 1.0,1.0);
```

```
glRotate(90, 0, 0, 1);
```

```
glPopMatrix();
```

```
drawHouse();
```

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:

(1 Point)

☐ (0,-2)

☐ (2,-2)

☐ (-2,2)

☐ (-2,-2)

☒ (2,2)

33

`diffuse_intensity = max(normal dot light_dir, 0)`

(1 Point)

☒ True

☐ False

34

`glNormal` can be called between `glBegin` and `glEnd`

(1 Point)

☒ True

☐ False

35

The projection command preserves straight lines

(1 Point)

☐ `gluPerspective`

☐ `gluLookat`

☒ `glOrtho`

☐ `glFrustum`

☐ `glViewport`

36

If the light position is $(1, 0, 0, 0)$, the attenuation factor at vertex $(0.5, 0.5, 1) =$

.....

(1 Point)

- ☐ Zero
- ☐ 0.5
- ☒ 1
- ☐ Cannot be calculated

37

In the Phong reflectance model, if only diffusion light component is used; a flat surface illuminated with a directional light source will have a uniform color intensity across the surface

(1 Point)

- ☒ True
- ☐ False

38

In OpenGL, lines color is set using____
(1 Point)

- ☒ glColor3f()
- ☐ glColor()
- ☐ glLineColor()
- ☐ None of these

39

What is the significance of i,f,v,d... in function naming conventions found in OpenGL?
(1 Point)

- ☐ OpenGL's naming conventions vary and must be examined in the specification for the version you are using
- ☒ Specify the expected parameter types (int, float, pointer, double, etc.)
- ☐ Specify the target type for variables written by OpenGL on the CPU-side only
- ☐ Specify the output target (immediate mode, frame buffer, virtual mode, double-buffer, etc.)

40

The inverse of a rotation matrix is another matrix such that when the two are multiplied together to get the_____.

(1 Point)

- ☐ Transpose the matrix
- ☒ Identity matrix
- ☐ Diagonal matrix
- ☐ None of these

41

If the rotation matrix R rotates a point by the angle θ , then the transpose of R rotates the point by $-\theta$

(1 Point)

- ☒ True
- ☐ False

42

`glTexCoord*()` is used for placing a text in a specific location in the window

(1 Point)

- ☐ True
- ☒ False

43

The process of displaying 3D into a 2D display unit is called as____
(1 Point)

- ☐ Resolution
- ☒ Projection ✓
- ☐ Rasterization
- ☐ Transformation

44

Once lighting is enabled, what happens to glColor() calls?
(1 Point)

- ☐ A run-time error occurs if both are used.
- ☒ They are ignored. ✓
- ☐ They function as normal.
- ☐ They enhance the shading effects.

45

An example of positional light source is the sun
(1 Point)

- ☐ True
- ☒ False

46

Is the following code, correct?

```
glBegin (LINES);
glVertex2i (1, 0);
glVertex2i (1, 1);
glVertex2i (-1, 1);
glVertex2i (-1, 0);
glEnd;
```

(1 Point)

☐ Yes

☒ No

47

Rotating the 3D point $(-1, -1, 0)$ by 180 around z-axis gives what:

(1 Point)

☐ $(1, -1, 0)$

☐ $(-1, 1, 0)$

☒ $(1, 1, 0)$

☐ $(0, -1, 0)$

☐ $(-1, -1, 0)$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

48

The plane is given by $2x+2y+z = 6$. What is the normal to the plane N?
(1 Point)

- ☒ (1, 1, 0.5) (2, 2, 1)
- ☐ (2, 2, 3)
- ☐ (2, 1, 0)
- ☐ (2, 0, 1)

49

When using the Phong Reflectance Model, we calculate the red, green, and blue color channels independently

(1 Point)

- ☒ True ✓
- ☐ False

50

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glScalef(-2.0, 2.0, 1.0);
glPushMatrix();
glTranslatef(-1.0f, -2.0f, -3.0f);
glRotatef(-90.0, 0, 0, 1.0);
glScalef (-1.0,0.5, -0.5);
glPopMatrix();
glRotatef(90.0, 0, 0, 1.0);
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -0.5 & 0 & 0 \\ 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 2 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1.5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 & -1.0 \\ 0 & -2 & 0 & -2.0 \\ 0 & 0 & 1 & -3.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☐ C

☒ D

51

OpenGL has no functions for keyboard input or mouse interaction

(1 Point)

☒ True

☐ False

52

..... light component comes from one direction and once it hits a surface, it is scattered.

(1 Point)

- ☒ Diffuse
- ☐ Specular
- ☐ Ambient
- ☐ Spot

53

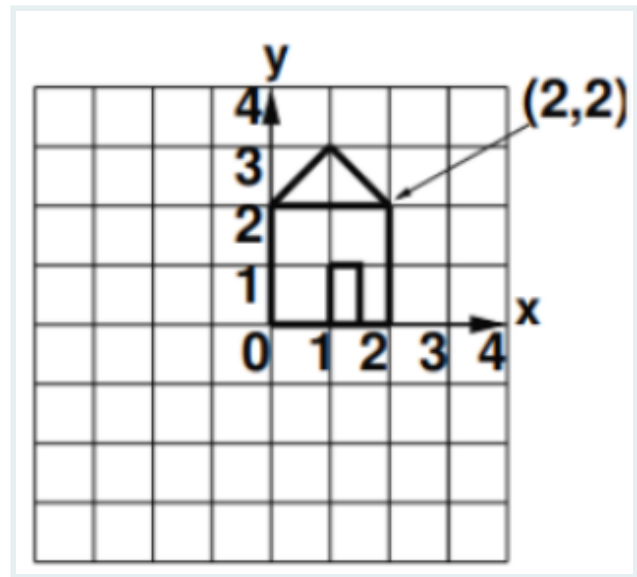
Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glScalef (-1.0,1.0, -1.0);
glScalef (1.0,0.5, 1.0);
(1 Point)
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0.5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0.5 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ☐ A
- ☐ B
- ☒ C
- ☐ D

54



Assume that drawHouse function draws the following house

```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0); glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glScalef(1.0, -1.0,1.0);
drawHouse();
glPopMatrix();
```

(2, -2)

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:

(1 Point)

- ☐ (0,-2)
- ☒ (2,-2) ✓
- ☐ (-2,2)
- ☐ (-2,-2)
- ☐ (2,2)

55

In glVertex2i() command, 2 represents _____
(1 Point)

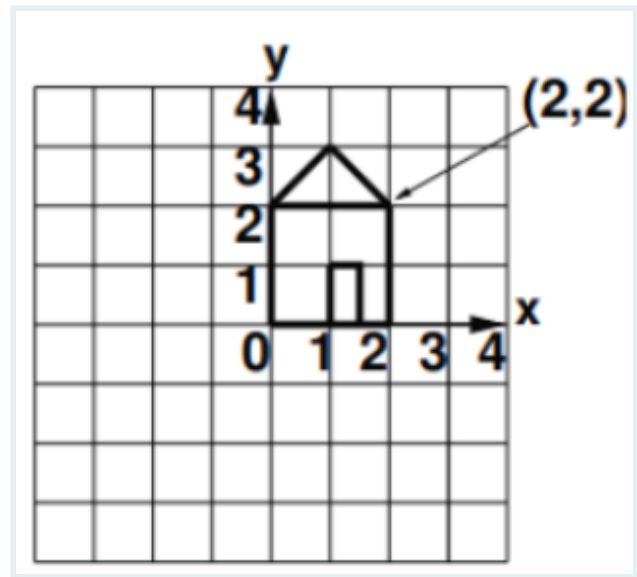
- ☐ type of arguments
- ☐ basic command
- ☒ number of arguments ✓
- ☐ None of these

56

diffusion light component is view independent
(1 Point)

- ☒ True ✓
- ☐ False

57



```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
```

```
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0); glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glPushMatrix();
```

```
glScalef(-1.0, 1.0, 1.0);
```

```
glRotate(90, 0, 0, 1);
```

```
drawHouse();
```

```
glPopMatrix();
```

$(2, 2)$
 $(-2, 2)$

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:

(1 Point)

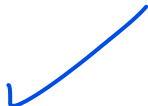
☐ (0,-2)

☐ (2,-2)

☐ (-2,2)

☐ (-2,-2)

☒ (2,2)



☐ Send me an email receipt of my responses

This content is created by the owner of the form. The data you submit will be sent to the form owner. Microsoft is not responsible for the privacy or security practices of its customers, including those of this form owner. Never give out your password.

Powered by Microsoft Forms | [Privacy and cookies \(https://go.microsoft.com/fwlink/p/?linkid=857875\)](https://go.microsoft.com/fwlink/p/?linkid=857875) | [Terms of use \(https://go.microsoft.com/fwlink/p/?LinkId=2083423\)](https://go.microsoft.com/fwlink/p/?LinkId=2083423)